

Proposal to form a SETI SCOPE (Society, Communications, Outreach, Post-Detection, Ethics) SubCommittee within the IAA SETI Permanent Committee

To: Director-General of the International Academy of Astronautics, IAA SETI Permanent Committee Executive, and IAA SETI Permanent Committee

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1. Summary

The SETI Permanent Committee of the International Academy of Astronautics (IAA) is a respected group of experts with specialist knowledge from many disciplines relevant to SETI. The Committee itself is evolving, and SETI science is rapidly changing and expanding, and so are the larger scientific and social landscapes. SETI has always had to contend with key questions, such as: How can signals be detected and confirmed? How should SETI scientists handle the scientific, political, and ethical responsibilities of a detection event? What should happen next? These questions have grown considerably more complex over time and require renewed and intensified efforts to address them comprehensively.

The SETI Permanent Committee has traditionally had a Post Detection Task Force or similar subcommittee. Given that there are at present no such subcommittees, **we propose a new SETI SCOPE subcommittee (Society, Communications, Outreach, Post-Detection, Ethics)** which will create a forward-looking strategy for the decade ahead. Such an active presence will also serve as a positive organisational capability for public engagement.

Ever since the formation of SETI, a Post-Detection capability has been required but often neglected. This capability will be necessary until SETI no longer exists. Arguably, a Post-Detection capability could and should exist outside of the Search, as we may give up on this (in the future) but 'they' may not give up on us; thus, we may have contact when no longer actively searching. We therefore submit that such a subcommittee should exist at least as long as the SETI committee does. In that way, Post-Detection expertise will have maintained and developed capabilities, so strategies will be in place, for action.

2. Rationale for Subcommittee Creation

The proliferation of exoplanet discoveries has increased societal awareness that humans may detect extraterrestrial life, including intelligent life. Global interest in the discovery of extraterrestrial intelligence has also intensified through new revelations in astrobiology, high profile technosignature searches, and increased computing and analytical capacity.

The IAA SETI Committee, established in 1974, is unique in that it remains the sole international long-term non-governmental body charged with examining all aspects of the search for extraterrestrial intelligence.

This includes an expectation that the IAA SETI Committee will lead discussions of the implications of detecting extraterrestrial intelligent life beyond earth.

During the 1980s the SETI Committee drafted “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”. The document was approved in 1989 by the IAA Board of Trustees and by the Board of Directors of the International Institute of Space Law. Although the Declaration and an accompanying position paper was presented to the United Nations Committee on the Peaceful Uses of Outer Space (COPUOS) in 2000, there remains no formal agreement among Earth’s nations in advance of a detection of extraterrestrial life.

The 2010 Declaration of Principles – unanimously adopted by the IAA SETI Committee in Prague in 2010, intended to supersede the Protocols – notes: http://resources.iaaseti.org/protocols_rev2010.pdf “#7 Post Detection.: *A Post-Detection Task Group under the auspices of the IAA SETI Permanent Study Group has been established to assist in matters that may arise in the event of a confirmed signal, and to support the scientific and public analysis by offering guidance, interpretation, and discussion of the wider implications of the detection.*” By not having a Post Detection Task Group since 2015, the IAA SETI Committee has been failing to follow its own Principles.

The detection of extraterrestrial intelligent life or its technology is anticipated to be one of the most important events in human history. The IAA SETI Committee is an established international scientific body with the collective ability to draw upon appropriate expertise to address both the technological and social impacts of a detection confirmation.

3. The Road Ahead

SETI has always had key practical dilemmas:

- How can evidence of ETI be found and confirmed?
- How should that information be shared with the scientific community?
- How should that information be shared with the global public?
- What should happen next? (further studies of the evidence, political decisions, communication efforts, etc.)

These key problems can be provisionally solved, but they cannot have *long-term* solutions, because SETI itself changes continually, so does the larger scientific landscape, and so does the world around us. Furthermore, the rate of change is accelerating; therefore, we must renew and redouble our efforts to keep SETI well-situated. Recent efforts by committee members (Rio 2.0 and the Protocols revision underway) are an excellent beginning, but there is much more to be done. The introduction of new collaborative working technologies within the Committee and quarterly virtual meetings, and the expanding Committee membership, has created an opportunity for innovation and renewal.

New variations of these old questions include these:

- What new arenas of astronomical *and other sciences* might bring ETI detections? How can those be potentially confirmed? What would the next phases of analysis be?
- How can we create and maintain an efficient 'handshake' between Detection and Post-Detection phases? e.g. discovered data handover; Rio Scale assessments to post detection analytics [DISC Quotient framework] and dissemination;
- How is a discovery process likely to unfold now? (e.g. a long delay between detection of a potential technosignature or exoplanet biosphere and follow-up studies; lengthy periods of contention or uncertainty);
- How should SETI and SETI Area Studies be communicated to the larger science community and to the interested international public, given the global communications and social media landscape?
- How can SETI researchers best meet their evolving ethical responsibilities?
- What are the *current* global concerns about societal impacts of potential or definite detections of, or contact with, extraterrestrial intelligence?
- What are the cumulative social effects of the absence, to date, of ETI detections?
- How will an astrobiology detection within our solar system, or the detection of an exoplanet biosphere, affect SETI?
- What Earth-based studies concerning the evolution of intelligence, social complexity, communication, language, and technology are needed to address questions which will arise upon the detection of LITU (Life in the Universe) or ETI (ExtraTerrestrial Intelligence)?
- How would a SETI detection of artificial/postbiological intelligence be received by the world?
- Where does SETI fit now in the rapidly evolving world of international space law and space policy?
- How should governmental science and privately-sponsored SETI initiatives relate? How do their challenges interfacing with the scientific community, and with the public, differ?

These questions require our focus. In addition to the expertise within the SETI Permanent Committee, many researchers worldwide are doing relevant work on potential types of detections of extraterrestrial intelligence (including but not limited to classic SETI), policy, the evolution and nature of intelligence and communication and social complexity, scientists' communication with the public, potential social impact of discoveries or contact, and modes of analysis for potential incoming signals. This committee will facilitate conversations with such researchers and research centers.

This SCOPE subcommittee will not address the questions and debates around *de novo* transmissions from Earth.

4. Proposed Initial Structure

We propose a small steering committee primarily composed of longstanding IAA SETI Permanent Committee members with relevant knowledge and extensive leadership experience, who will undertake

the task of creating a new, inclusive SCOPE subcommittee and framework which allows contributions from all interested IAA SETI Committee members, including newcomers.

Initial Steering Group Members: Paul Davies, John Elliott, Denise Herzing, Kathryn Denning, Lori Walton, Les Tennen, Duncan Forgan, Carol Oliver , XXXXXXXX, XXXXXXXX, XXXXXXXX. [Three additional people are also interested. Awaiting confirmation from them.]

The final SETI SCOPE subcommittee structure, to be announced in summer 2019, will depend on which IAA SETI members wish to be involved. We anticipate that the final SETI SCOPE subcommittee will have Leads for:

- the group's organization/coordination and resource library
- research areas / subject specialists: information sciences, social sciences, biological sciences, astrobiology, astronomy, humanities, law and metalaw, space policy, and communications scholarship concerning social media and news
- next generation capacity-building (i.e. encouraging universities to offer relevant training, encouraging post-graduate students to develop relevant expertise, and facilitating cross-training for professionals to learn about each others' fields)
- liaison with related research centers worldwide

5. Proposed Initial Schedule for SETI SCOPE Subcommittee – or SCOPE Temporary Working Group

Spring 2018 – Summer 2019:

- We will consult within the IAA SETI committee and the larger interested community, to compile information, identify gaps and prepare a draft framework for future work, for a subcommittee which is sustainable over the long term.
- We will consult widely about areas within SCOPE which require discussion, research, and knowledge mobilization.
- We will work together with committee members who have been revising the Rio Scale and the Protocols.
- We will offer briefing documents/presentations about the history and current landscape of postdetection/SCOPE concerns for new/junior SETI Permanent Committee members, so they can rapidly become fully informed participants in these complex conversations.

Details will depend upon resources which can be mustered (e.g. for website hosting, teleconferencing, in-person workshops, etc.), but the aim is to be inclusive, including people who cannot regularly travel, attend expensive overseas meetings, etc.

Committee Meeting at IAC 2019, Washington DC, 21-25 October 2019:

- We will present a new framework / strategy document for this SETI SCOPE Subcommittee. We will provide a written paper for the committee members who are unable to attend the IA Congress.

Special Note: This new subcommittee will be open to any interested IAA SETI Permanent Committee members who have relevant expertise. We invite those who are interested to identify themselves, so we can start building a common conversation which will take us through the next decade. Please send an email to John Elliott at setijon@aol.com and Kathryn Denning at kdenning@yorku.ca with a paragraph about your pertinent interests, a list of any specifically relevant publications and/or current research or policy endeavours, and a summary of your IAA SETI membership history and prior activities (if any) within the IAA SETI group and within the larger IAA. With this information, we can start creating a mutually aware network of participants. We intend to regularly report to and seek input from the larger IAA SETI Committee.